



BIT1123/BISE2093/DIT1113  
BIT1123 OBJECT ORIENTED PROGRAMMING

ASSIGNMENT 1 – INDIVIDUAL  
(20%)

| No. | Name                      | ID           | PROGRAM |
|-----|---------------------------|--------------|---------|
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ALL SECTIONS

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## 1. Introduction

This reflective report depicts my learning experience while completing the Object-Oriented Programming (OOP) tutorials in the Java programming language. The tutorials covered the basic programming concepts of Java and increased its complexity with the introduction of more advanced concepts of OOP.

In the tutorials, I was writing java programs with graphical user interface, abstract classes, classes and objects, methods, attributes and constructors, and inheritance. I also got to know how to structure a programming project on GitHub, and how to structure the source code in a project repository.

The principal focus of this learning journey would have been not just to learn the syntax of Java but how Object Oriented Programming can be used to design and organise software programs. The tutorials and their maintenance into one GitHub repository helped me to see the progress made from simple Java to more structured OOP-based applications as I went.

The tutorial code and files for the course are contained in my GitHub repository. It also shows my experience in organizing, documenting and version controlling a repository.

## 2. The following is a list of topics covered in each tutorial.

This will help you get started as a new programmer. This is a first week of classes, objects, and basic java.

The first tutorial was an introduction to the basic class and object concepts of the java programming language. I learned how to write a java class and how to define attributes which describe an object.

For instance, the Student class had attributes like name, age, and GPA. I also learned how to create an object from a class and how to access the methods of the class.

This tutorial helped me to understand the basic relationship between a class and an object. Classes can be thought of as blueprints for objects, which are instances of the class.

This week, we studied classes and objects in practice. This week, we learnt about classes and objects in practice.

In week 2, I furthered my practice of creating and manipulating Java classes and objects. I got a lot of practice declaring attributes, creating objects, and calling methods during the exercises.

This tutorial helped me to understand, how different parts of a java program communicate with each other. I also got a better understanding of the organization of Java programs and how the main method is used to run a program.

### Week 3–4 – Inheritance and Relationships Between Classes

In Weeks 3 and 4 there were other more significant OOP concepts introduced, specifically relationships between classes and inheritance.

There are classes like Person, Student, Lecturer and a class called Main in the repository. It made me realise how classes can be related to one another and that common characteristics can be given to the parent class and specialized classes can derive the common characteristics.

I've discovered that inheritance increases program structure and decreases program duplication. It also gave me an idea of what parent and child class are in OOP.

## Week 5 – OOP Practice and Documentation

In week 5 there was more practice using the concepts of Object Oriented Programming. It is comprised of java source codes and documentation.

This tutorial really taught me that programming is about more than coding. Documentation and organization is also key, as it helps to make the program easier to understand, maintain, and review.

I also gained a greater understanding about the need to employ meaningful organization of files and classes when programming assignments.

This week we will be covering Employee and Lecturer Classes. This week we will focus on Employee and Lecturer Classes.

I was involved with employee classes and lecturer classes in Week 6. The repository has `employee.java`, `lecturer.java` and `main.java`.

This exercise was to further practice inheritance and class relationship. I learned that a child class can inherit from a parent class and give more specific characteristics or behaviours to the parent class.

The experience of using these classes was a great way to learn about how OOP can model real world relationships in software. For instance, a lecturer can be defined as a special category of an employee.

In this week's class, we will explore abstract classes and look at an example that involves appliances.

The concept of classes related to appliances was introduced in week 7, which was an important OOP concept. The repository has Appliance, Microwave, WashingMachine and Main.

This exercise made me realize that the class GeneralParent can be used to represent some common properties, and the subclasses like ChildA and ChildB can be used to represent the specific types of objects.

I became more familiar with abstraction and inheritance as well. A program can be organized into multiple classes, rather than a single large class, into which all the functionality is packaged.

Learn fundamental programming concepts in Java through hands-on exercises. Practice programming with JAVA by doing hands-on.

Week 8 and Week 9 were more of an application of Java programming concepts. There is a task.txt file and a main Java program in the repository.

These tutorials helped me to refresh my memory of the things I learnt in the previous weeks. Gained more familiarity in writing, testing and structuring Java source code.

I was also able to see in the practical work that programming can be improved by repetition and problem solving. After some uses, concepts that at first appeared challenging were easier.

## Week 10 – Java GUI Quiz Application

In tutorial week 10, graphical user interface (GUI) development was introduced and it was one of the most interesting weeks.

Presentations in this repository include the files: Questions.java and QuizBattleGUI.java. This project shows a small application that is interactive in nature and based on the quiz concept and implemented using java GUI programming.

This tutorial was helpful to me in being able to write several programming concepts in one at the same time instead of just simple console programs. I discovered that GUI applications need user interface components to interact with each other, to handle events, to use classes and to utilize program logic.

I also learned about the interactions that can be made in the development of the quiz application, as opposed to the other majority of software that are just command line programs.

### 3. Challenges Encountered

The most difficult part of the tutorials was getting a grasp of the concepts of classes and objects. Initially it was hard to see why a class needed to be written first to define its attributes and methods, and how the object would be able to call these.

Another difficulty was to grasp the concept of inheritance. It was initially confusing how parent and child classes were related, particularly to see which attributes and methods belonged to which class.

Also I had some problems when I worked with more than one java file. All classes had to be identified correctly and files had to cooperate appropriately. It is possible that a small error, in the class name, the file name, the name of the constructors or methods, may lead to compilation errors.

The GUI tutorial added more challenges than the previous console tuts, due to the addition of more components and interactions.

I also needed to work on a few other things, such as GitHub and repository organization. Careful organization was needed to manage several tutorial folders, source files, README documentation and a final report.

### 4. How the Challenges were overcome

I worked through these problems primarily through a process of trial and error, and constant testing. I did not just read examples, but changed the Java programs and noticed the changes of the results.

In classes/Objects: I practised making multiple objects from the same class, and calling different methods of the objects. This helped me to see the distinction between a class definition and an actual object.

I learned about inheritance as the relation between parent and child classes and practiced making an instance like that of Employee and Lecturer object. This made it easier to understand, as the relationship was a real-life scenario.

When compiling, if there were any errors, I would learn to pay attention to error messages and pinpoint the class, method, or line that caused the error. I also discovered that Java has a case sensitivity and capitalization.

In the GUI application, I took a step by step approach to the program. I started by trying to figure out what classes were used and then explored the relationship between the graphical elements and actions.

Lastly, I became familiar with the need for structuring projects in a clear and organized manner and organizing source code, thanks to my regular use of GitHub.

## 5. Enhancement of my Java programming skills

My programming capabilities with java improved greatly in these tutorials. Initially, I was concerned with learning the basic syntax and creating simple programs.

Later I got more comfortable to create multiple classes and to link them together. I enhanced my skills of defining attributes, constructors, methods and objects.

I also learned more about inheritance. Increased ability to recognize common characteristics and determine when classes should have parent child relationships.

The other thing that improved is my ability to debug. When the program didn't compile or run as expected, I developed more faith in the ability to check the source code and find errors.



I also gained some new skills in organizing java projects. I learnt to create folders for the tutorial work.

My last GUI project also helped me with the understanding of how to do application development since I had to have both the logic and the user interaction.

Ability to comprehend Object Oriented Programming concepts.

The tutorials gave me a better understanding of the fundamental concepts of object oriented programming.

### Classes and Objects

A class is a blueprint of an object, that determines its structure and behaviour. Classes define values for objects, which can be created from the classes and contain their own values for the attributes.

### Encapsulation

Encapsulation is the ability to make data and associated methods part of a class. This offers a means of controlling information and actions.

### Inheritance

One class can inherit characteristics and behaviours from another class. This relationship is very evident from the Employee and Lecturer examples that I took.

### Polymorphism

Polymorphism enables objects related by inheritance to be handled flexibly. The concept of inheritance and related classes was used as the basis to understand it.

### Abstraction

I grasped this concept of how a general class can be used to represent common characteristics and more specific child classes can be used for more specific implementations with the appliance example.

Overall, I find that OOP is beneficial because it enables software to be broken down into logical and reusable parts rather than writing just one big program.

What is Version Control and how do you use it? What is Version Control and How to use it on GitHub?

Another significant component of this assignment was to learn how to use GitHub as a repository and software development management system.

I created one repository that includes the work of the Java tutorial and support files. It has separate folders for each week for easier reference and review.

A README file with information about the course, the contents of the tutorial, the technologies used, the concepts practiced and how to execute the programs is also included in the repository.

I learned the significance of version control in software development from the use of GitHub. It offers a single repository of source code and it enables users to trace changes made to the source code over time.

This experience will be helpful for any software development project in the future, since Git and GitHub are tools that are used for managing source code and collaborating with fellow programmers on programming projects.

## 8. Future Learning Plans

In the future, I will continue to work on my java skills, by creating larger applications instead of just smaller ones in tutorials.

I would like to do more advanced OOP (Object Oriented Programming) practice on interfaces, encapsulation, polymorphism, exception handling, collections, etc.

I am also going to learn to develop a GUI using the GUI toolkit and design apps with improved user interface and functionality.

One of the objectives is to get better at Git and GitHub. I wish to use branches, commits, pull requests, and improve my documentation of my repository in future projects.

Lastly, I will be integrating knowledge of Java and OOP with other areas of computer science to create full software applications with well structured, maintainable code.

## 9. Conclusion

Overall, I felt the Object Oriented Programming tutorials helped me get some hands-on experience with java. The learning experience began focusing on basic classes and objects and then moved on to inheritance, abstraction, class relationships and GUI development.

I found the tutorials useful in addition to just learning how to write Java code, they also helped me understand how a program can be structured using the principles of Object Oriented Programming. I was able to learn about how classes can be used to model real-life things and how objects of one type can be related to objects of another type using inheritance relationships.

I also acquired skills for debugging, better project organization, documentation and GitHub skills. During the tutorials i learned the difficulties involved in the tutorials which made me more patient and systematic when trying to solve a programming problem.

The last GUI Quiz app. showed how all the concepts learned in the tutorials can be integrated into a more interactive java program.

Overall, this assignment has given me a lot of confidence in java and has given me a firm base for learning more advanced java skills and concepts in software development in the future.

## 10.GitHub Repository URL

<https://github.com/Mohammeddeya/MOHAMED-DEYAELDIN-202505010510/tree/main>